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Effect of moisture on the yield locus of granular materials: theory of shift

Patrice Pierrat, Dilip K. Agrawal, Hugo S. Caram *

Lehigh University, Chemical Engineering Department, Iacocca Hall, 111 Research Drive, Bethlehem, PA 18015, USA

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Abstract

The yield locus of a wet material was shifted to the left of that of a dry specimen by a constant value equal to the compressive isostatic stress due to pendular bridges. For materials with straight yield loci, the shift was computed from the uniaxial tensile strength, either measured in a tensile strength tester or calculated from the correlation, and the angle of internal friction of the material. The predicted shift in the yield loci due to different moisture contents compare well with the measured shift in the yield loci of glass beads, crushed limestone, super D catalyst and Leslie coal.

Keywords: Characterization; Wet solids; Yield locus; Tensile strength

1. Introduction

It is well known that the wet granular materials do not flow well. From sand castles to hang up coal bins, evidences are all around. The 'stickiness' of the wet solids is commonly attributed to capillary forces between the particles. Capillary forces are reasonably well understood and can be related to the tensile strength of the material [1]. On the other hand, the flow properties of a granular material are commonly determined experimentally through shear testing. According to Jenike [2], a significant number of tests must be performed to determine the flow properties of a material. These flow properties depend on several parameters such as the size, size distribution and shape of the particles, the composition of the granule and its moisture content. A good theory can significantly reduce the number of measurements required and provide guidance on the parameters affecting the flow properties. Some effort has been reported earlier for the analysis of flow of cohesive powders where van der Waals forces are important but not for the wet materials [3].

The objective of this paper is to provide a fundamental connection between the yield locus of a dry material (free flowing), the tensile strength of the wet material and the yield locus of the wet material.

2. Theory of shift

The yield curve of a granular material can be seen as the envelope of the Mohr circles describing the state of stress of the material at yield. Construction of the yield curve allows for the determination of the Mohr circles as is shown in Fig. 1. The dotted Mohr circles represent the state of stress at each condition from the point of view of an external observer accounting for the external forces applied to the system.

In the case of wet granular materials the presence of moisture introduces capillary forces. If there is no geometric ordering of the system, Rumpf [4] has shown that these capillary forces introduce an isotropic compressive stress or isostatic tensile stress (ITS) into the system.

The *true* state of stress of the granular material will then be in each case represented by a Mohr circle shifted to the right with both principal stresses increased by the value of the ITS. An example of this shift is shown by a full line circle in Fig. 1. All of the Mohr circles tangent to the yield curve will be displaced by the same amount.

The *theory of shift* assumes that the intrinsic yield locus of the wet material is the same as the intrinsic yield locus of the

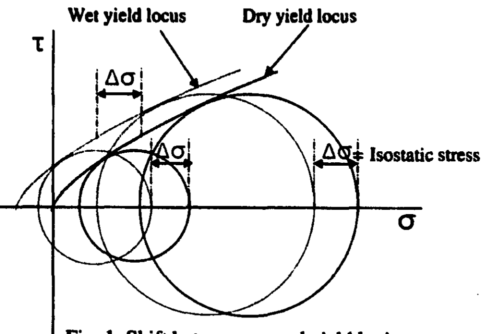

Fig. 1. Shift between curved yield loci.

* Corresponding author. Tel.: +1-601-758-4259; Fax: +1-610-758-5057; E-mail: hsc0@le4high.edu

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dry material. The intrinsic yield locus can be defined as a yield locus of the material when all of the stresses are accounted for. Differences between the observed yield loci of dry and wet materials can be eliminated if the ITS is included in the analysis. The net effect is that the observed yield curve is shifted to the right by a constant amount equal to the ITS.

The value of the ITS is not easily measured but can be estimated by measuring the uniaxial tensile strength of a sample of the same void fraction. In the simplest case the yield locus can be assumed to be a straight line that can be extrapolated into the tensile region of the shear-normal stress diagram. This is shown in Fig. 2. The isostatic stress will be represented by the point F in the diagram. The uniaxial tensile strength ΔT is given by the point E, obtained drawing the Mohr circle tangent to the yield locus and passing through the origin.

When the yield locus is approximated by a straight line, the following relation for the separation distance between the yield locus can be derived from the geometry of Fig. 2:

$$\Delta\sigma=\Delta T\frac{1+\sin\phi}{2\sin\phi} \tag{1}$$

To sum up, when moisture is added to a dry powder, the yield locus of the wet powder is parallel to the dry yield locus and shifted to its left by a value depending on the tensile strength of the wet powder and the angle of internal friction of the material. This value, equal to the isostatic tensile strength, is given by Eq. (1) where ΔT is the uniaxial tensile strength. The validity of the above theory must be verified experimentally before it can be used to obtain the yield loci of powders at different moisture contents from the sole knowledge of one yield locus and the tensile strength of the powder.

The reader will find some similarities between the Warren Spring model and the model presented but with different physical and mathematical approach. In particular, the Warren Spring model assumes uniaxial tensile to be identical to isostatic tensile strength (ITS). No such assumption is made in this model. With this provision, our model corresponds to the Warren Spring model with $n=1$.

$$\left(\frac{\tau}{\text{ITS}\tan\phi}\right)^n=\frac{\sigma+T}{T} \tag{2}$$

3. Experimental work

Tests were performed on five dry granular materials, glass beads [1], 93 μm and 180 μm in diameter, super D catalyst [2], 104 μm in diameter, crushed limestone [3] with sieve size 8–180 μm and coal [4] with sieve size 88–212 μm in order to determine the yield locus of the dry material. Experiments were also conducted on samples at different moisture contents. The complete yield loci were obtained by combining data from a modified shear apparatus described below and tensile strength measurements obtained using a Parfitt tensile tester described in detail elsewhere [1].

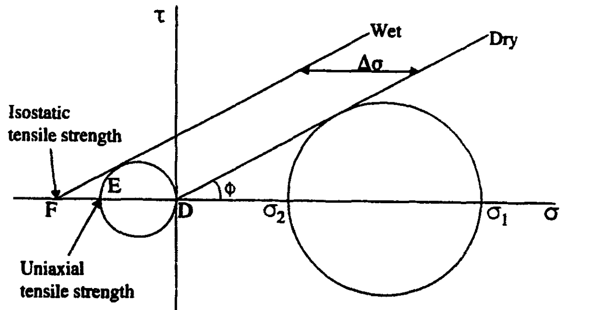

Fig. 2. Yield loci for the dry and wet material corresponding to the failure lines drawn in the plane of principal stresses.

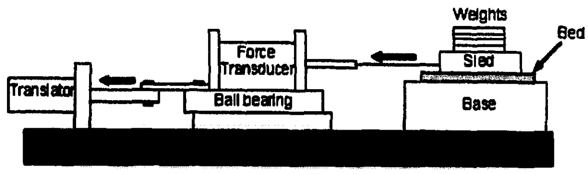

Fig. 3. Schematic representation of the modified shear tester.

3.1. Shear tests

The tests were performed on the modified shear apparatus similar to that described by Hiestand et al. [5] for the shear testing on pharmaceutical powders. A schematic representation of the tester is given in Fig. 3. A thin bed 3.1 mm. thick of the powder was laid on a fixed base. The powder was scraped with a spatula to level the bed while avoiding packing it, and a sled was gently placed on top of the bed. The bottom of the sled and the top of the base were grooved in order to prevent slip at the contact surfaces and to ensure that a shear plane would develop within the bulk. There were 11 grooves per centimeter and each groove was 1.0 mm deep. The sled was 4.8 cm in length and 3.4 cm in width. It was assumed that the top layer of particles would interlock with the grooves and that friction or shear would occur on a plane away from the top. Weights were applied on the top of the sled which was pulled by a fishing line string attached to an inchworm translator. Positioned between the translator and the sled was a force transducer resting upon a set of low friction ball bearings. The transducer was connected to a personal computer where the shear force was recorded as a function of time. The testing procedure included 3 steps. First, the powder was laid on top of the base to form a flat thin bed of loosely packed particles. The sled was positioned on the bed and a

[1] Potter Industries, Cleveland, OH 44113, USA.

[2] W.R. Grace and Co., Davison Chemical Division, Cincinnati, OH 45229, USA.

[3] AGWAY, Syracuse, NY 13221, USA.

[4] Leslie mines in Western Pennsylvania provided by New York State Electric and Gas.

normal load was applied on top. Second, the inchworm translator was turned on and started pulling the sled. Consolidation was accomplished by causing the specimen to flow with shear until a steady state value of the recorded shear stress was closely attained. Finally, when consolidation was completed, the force was released by switching the inchworm translator into the reverse mode. The vertical load was then replaced by a smaller load and the specimen was sheared. This procedure was repeated 5 times to provide 5 points on the yield locus. Each test was replicated 3 times and an average value of the recorded shear was used for the determination of the yield locus.

3.2. Tensile strength tests

Tensile strength tests were carried out using a Parfitt tensile tester with an enlarged split cell (inside diameter 81 mm, height 25 mm) as described in Ref. [6]. It was first necessary to determine the bulk density of the specimens before shearing in order to conduct tensile strength tests at the same packing density. The procedure was as follows. The sample was consolidated using the modified shear tester. The volume of the sample was determined by measuring the average thickness of the bed. The sample was weighed with the base so the mass of the powder could be calculated. From the volume of the sample and its mass, the bulk density was calculated and is reported in Table 1. For tensile strength measurement, a known quantity of the powder was put into the split cell. In order to obtain the same packing density as in the shear test, the powder was compacted to a certain bed height. Once the sample was compacted, the cell was fastened on the Parfitt tensile tester [7] and a tensile test was performed. The procedure was repeated 4 times to minimize the error. The measured values of the tensile strength of crushed limestone, super D catalyst and Leslie coal at different moisture contents are reported in Table 2 along with their 95% confidence interval. Tensile strengths of glass beads were calculated using the correlation proposed by Pierrat and Caram [1] and are reported in Table 3. The correlation allows for the calculation of the tensile strength of a powder in the plateau region of the pendular state from the knowledge of parameters which are easy to measure. The tensile strength of powders with moisture content less than that corresponding to the plateau region can be extrapolated if one assumes that it is proportional to the bonding force between particles. For the dimensionless separation distance, the value $a/d = 0.035$ was used in the paper.

Table 1
Bulk density of the powders

Material	Wetting solution	Moisture content (%)	Bulk density (g/cm^3)
Glass beads (93 μm)	Water	1.3	1.459±0.030
	Water	4.0	1.534±0.019
	n-hexadecane	1.0	1.569±0.031
Glass beads (180 μm)	Water	1.3	1.435±0.003
	Water	4.0	1.497±0.008
	n-hexadecane	1.0	1.519±0.019
Crushed limestone	Water	1.3	1.288±0.013
	Water	4.0	1.329±0.021
	Water	10.0	1.387±0.024
Super D catalyst	Water	15.0	0.930±0.014
	Water	20.0	0.953±0.012
	Water	25.0	1.000±0.028
Leslie coal	Water	4.0	0.659±0.011
	Water	10.0	0.684±0.014

Table 2
Measured values of the tensile strength of the various materials

Material	Wetting solution	Moisture content (%)	Tensile strength (g/cm^2)
Crushed limestone	Water	1.3	2.55±0.88
	Water	4.0	7.02±1.06
	Water	10.0	9.48±0.97
Super D catalyst	Water	15.0	2.60±1.24
	Water	20.0	9.13±1.18
	Water	25.0	10.90±0.95
Leslie coal	Water	4.0	2.31±1.14
	Water	10.0	7.81±1.29

Table 3
Calculated values of the tensile strength of the glass beads

Material	Wetting solution	Moisture content (%)	Tensile strength (g/cm^2)
Glass beads (93 μm)	Water	1.3	9.43
	Water	4.0	12.84
	n-hexadecane	1.0	4.19
Glass beads (180 μm)	Water	1.3	4.61
	Water	4.0	6.15
	n-hexadecane	1.0	1.99

4. Results

4.1. Measurements on glass beads

The yield loci for the dry and the wet material were drawn on the same graph. It was found that, for the dry glass beads, consolidation had no significant effect on the shear data (Fig. 4). But the consolidation had modest effect on the yield loci of the wet glass beads. Figs. 5 and 6 show the yield loci for the glass beads samples wetted with water and *n*-hexadecane at a consolidation pressure of 28.8 g/cm^2. The yield loci are nearly parallel to each other and the horizontal distance between them is almost constant over the entire range of normal stress. As expected, the yield locus of the dry glass

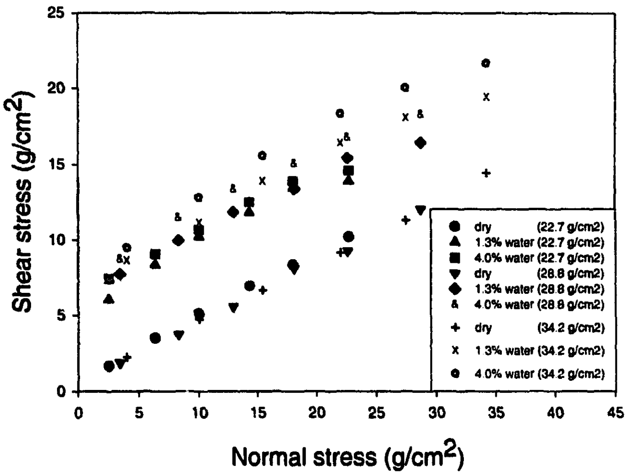

Fig. 4. Effect of consolidation pressure on the yield locus of glass beads (93 μm).

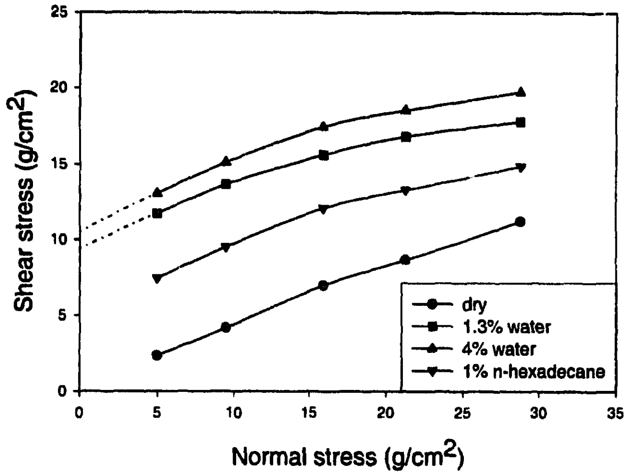

Fig. 5. Dry and wet yield loci of glass beads (93 μm).

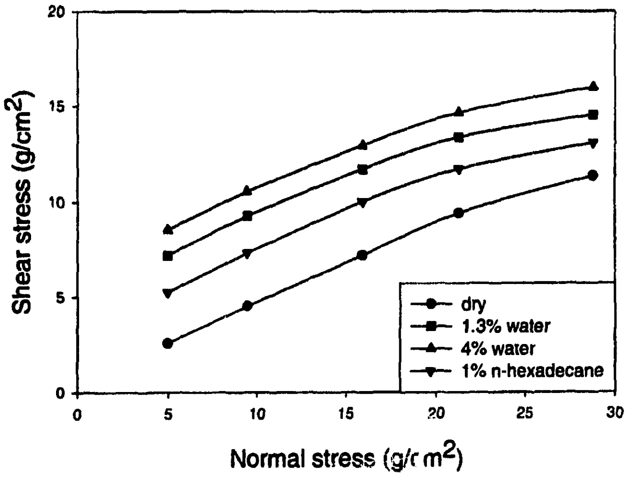

Fig. 6. Dry and wet yield of glass beads (180 μm).

beads lies below the wet yield loci and passes through the origin indicating no cohesion within the bulk. Also, the separation distance between the yield loci is larger for samples wetted with water than for samples wetted with n-hexadecane. This is also expected since the overall strength of the latter specimens is much less than the strength of the former ones and, consequently, variations due to difference in moisture content should be less also. The horizontal separation distance between the dry locus and the wet loci, as well as the one between the wet loci, was measured in the region of positive normal stress. Several measurements were made to cover the entire range available. The limitation was the dry yield locus which did not extend far enough to the right to allow for the measurement of the shift at high constant shear stresses. In the case of glass beads, 93 μm in diameter, yield loci of the samples wetted with water were extrapolated in the region of lower normal stress region (dotted lines in the Fig. 5) to give the shift between the yield loci of the dry sample and the samples wetted with water. Measurements of the separation distance between the wet loci were done over a wider range. In order to compare the measured distance with the one predicted by the theory described previously, it was necessary to assume that relation (1), derived for linear yield loci, could be used without inducing too much error since the curvature of the yield loci is very small. Relation (1) requires the knowledge of Φ, the angle of internal friction and the uniaxial tensile strength. Tensile strengths for glass beads were calculated as described earlier. Again, since the curvature of the yield loci was small, Φ was taken as constant and an average value was calculated. The slope of each yield locus considered was determined at a median value of the normal stress and averaged to give Φ (Table 4). The predicted shifts are reported and compared with the measured one in Table 5. Table 5 shows that, for samples wetted with both water and n-hexadecane, the measured separation distances between the yield loci of the dry and the wet specimens compare well with the predicted values of the shift. For easier analysis, the measured separation distance was plotted against the predicted shift. Fig. 7 shows plots for samples wetted with water and n-hexadecane. The line through the origin represents the locus where the measured shifts and the predicted ones are equal. The data points lie relatively close to the line. Measured and predicted shifts of the yield loci of glass beads of 180 μm diameter is almost half of the corresponding values for glass beads of 93 μm diameter. This was expected since the tensile strength of the latter specimens is almost double of the corresponding value for glass beads of 180 μm diameter and both materials have almost the same angle of internal friction. However, the measured shift is larger than the one predicted by relation (1). This may be explained as follows.

Table 4
Angle of internal friction of the various materials

Material	Average angle of internal friction (deg) (at a median normal stress)
Glass beads (93 μm)	16.1
Glass beads (180 μm)	19.8
Crushed limestone	29.3
Super D catalyst	20.5
Leslie coal	16.1

Table 5
Measured and predicted shift between the yield loci of the various powers wetted with water and n-hexadecane

		Measured shift (g/cm^2)	Predicted shift (g/cm^2)
Glass beads (93 μm)	Dry–1.3% water	26.28	21.77
	Dry–4.0% water	29.63	29.64
	Dry–1.0% n-hexadecane	14.01	9.67
	1.3% water–4.0% water	5.52	7.87
	1.0% n-hexadecane–1.3% water	12.49	12.10
	1.0% n-hexadecane–4.0% water	17.13	19.97
Glass beads (93 μm)	Dry–1.3% water	11.78	9.11
	Dry–4.0% water	15.22	12.16
	Dry–1.0% n-hexadecane	7.00	3.93
	1.3% water–4.0% water	3.91	3.04
	1.0% n-hexadecane–1.3% water	5.05	5.18
	1.0% n-hexadecane–4.0% water	8.60	8.22
Crushed limestone	Dry–1.3% water	4.00	3.88
	Dry–4.0% water	9.75	10.68
	Dry–10.0% water	15.56	14.42
	1.3% water–4.0% water	5.79	6.79
	1.3% water–10.0% water	12.55	10.53
	4.0% water–10.0% water	6.49	3.74
Super D catalyst	Dry–15.0% water	4.45	5.01
	Dry–20.0% water	19.28	17.60
	Dry–25.0% water	22.78	21.20
	15.0% water–20.0% water	14.92	12.58
	15.0% water–25.0% water	19.05	15.99
	20.0% water–25.0% water	4.64	3.41
Leslie coal	Dry–4.0% water	7.31	5.31
	Dry–10.0% water	19.02	17.94
	4.0% water–10.0% water	11.32	12.63

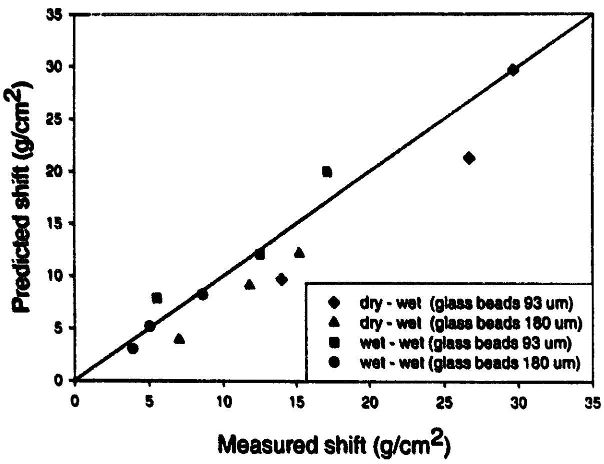

Fig. 7. Measured shift vs. predicted shift between yield loci for samples wetted with water and n-hexadecane.

For most of the material considered, the yield loci become flatter near the consolidation point. This region gives higher values of the measured shift than the predicted shift based on the angle of internal friction, Φ, at a median value of the normal stress. Finally, it is important to have accurate values of the tensile strength for the specimens considered so that a precise value of the difference in tensile strengths, ΔT, can be used in Eq. (1) to predict the shift.

4.2. Measurements on crushed limestone, super D catalyst and Leslie coal

Shear data on three other powders, crushed limestone, super D catalyst and Leslie coal were also used to verify the validity of the theory of shift. Samples were prepared by adding 1.3%, 4.0% and 10.0% distilled water to crushed limestone, 15.0%, 20.0% and 25.0% distilled water to super D catalyst and 4.0% and 10.0% distilled water to Leslie coal as received. The samples were consolidated at a pressure of 28.8 g/cm^2. Figs. 8–10 show the yield loci of the dry and the wet samples of crushed limestone, super D catalyst and Leslie coal respectively. Tensile strength tests were performed on crushed limestone, super D catalyst and Leslie coal at the same packing density as in the shear test described earlier. In most cases, the shift was found to be quite constant over the entire range of normal stresses. For Leslie coal, the curvature was more pronounced at the low values of the normal stress. The separation distance between the dry yield locus and the wet yield loci was measured and Eq. (1) was used to predict the shift. Both values are reported together on Fig. 11 for comparison sake. The line drawn is where the measured and the predicted shift are equal and the agreement is fairly good.

The theory of shift assumes that the intrinsic yield locus of the wet material is the same as the intrinsic yield locus of the dry material. To verify this, the intrinsic yield loci of the

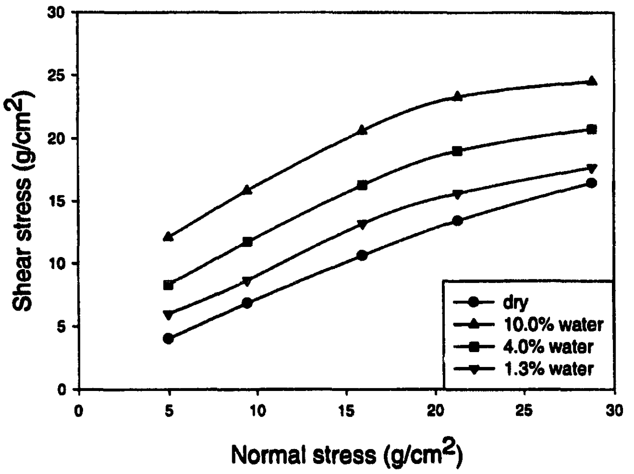

Fig. 8. Dry and wet yield loci of crushed limestone.

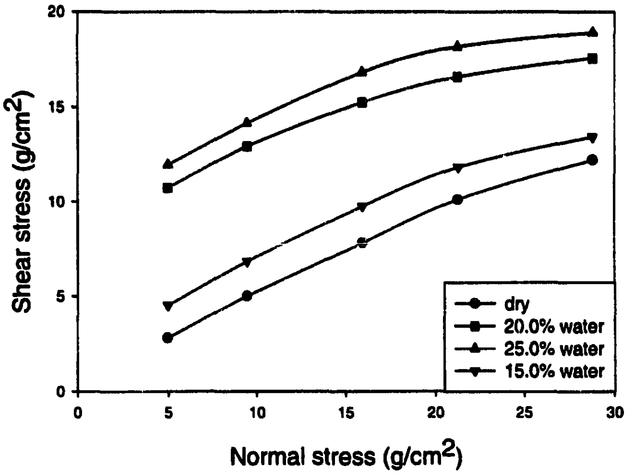

Fig. 9. Dry and wet yield loci of super D catalyst.

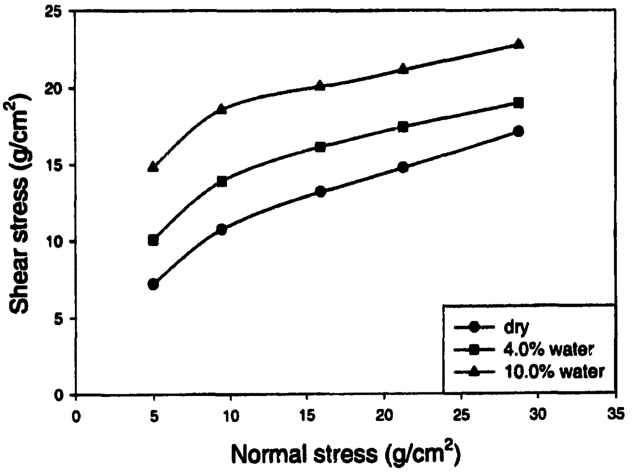

Fig. 10. Dry and wet yield loci of Leslie coal.

materials were obtained moving the wet yield loci to the right by the value of predicted shift between the wet and the dry yield loci. Figs. 12–16 show the intrinsic yield loci of the five bulk solids. The intrinsic yield loci of the wet material fall on the same line as the intrinsic yield locus of the dry material. This also strengthens the shift theory that the shift in yield

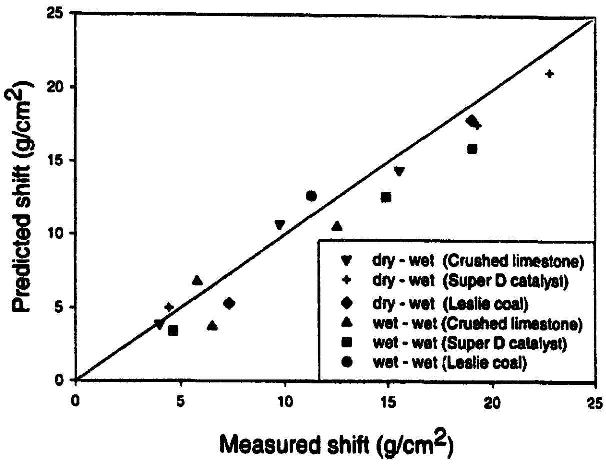

Fig. 11. Measured shift vs. predicted shift between yield loci for samples wetted with water.

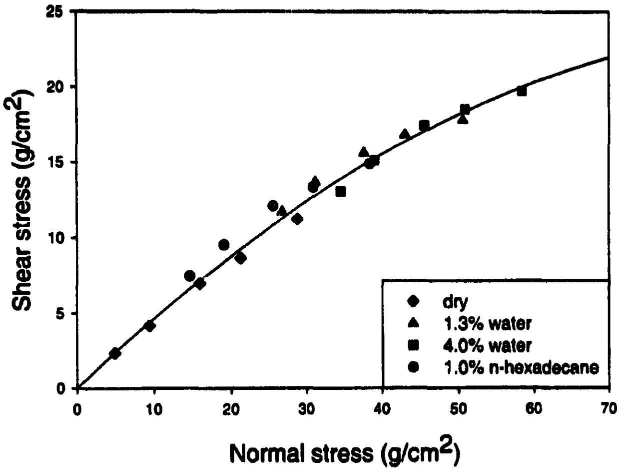

Fig. 12. Intrinsic yield locus of glass beads (93 μm).

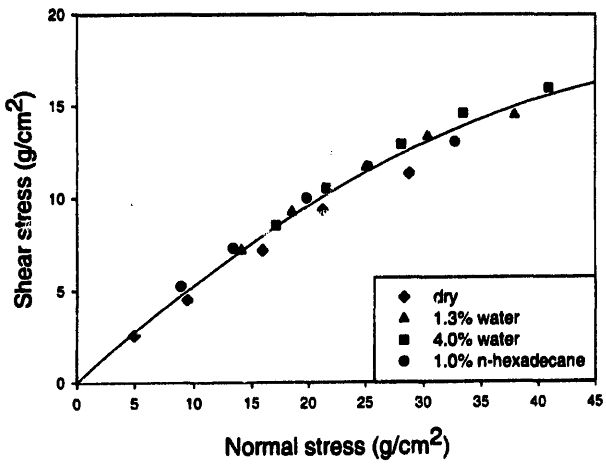

Fig. 13. Intrinsic yield locus of glass beads (180 μm).

loci of the wet material from the dry material is, caused by and is equal to, the isostatic tensile strength, introduced by the liquid bridges.

It is more difficult to predict the shift in the case of curved yield loci and relation (1) can only give an approximation

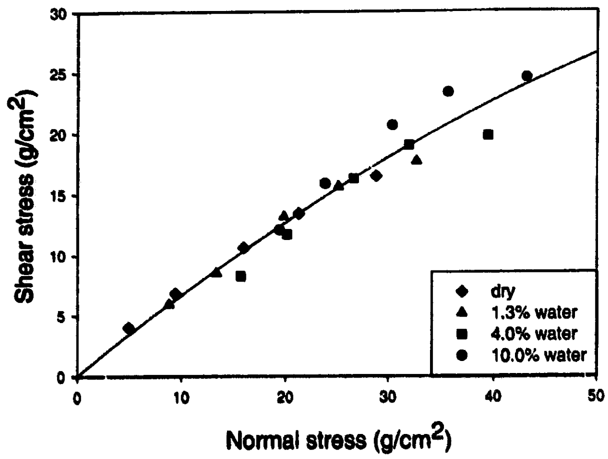

Fig. 14. Intrinsic yield locus of crushed limestone.

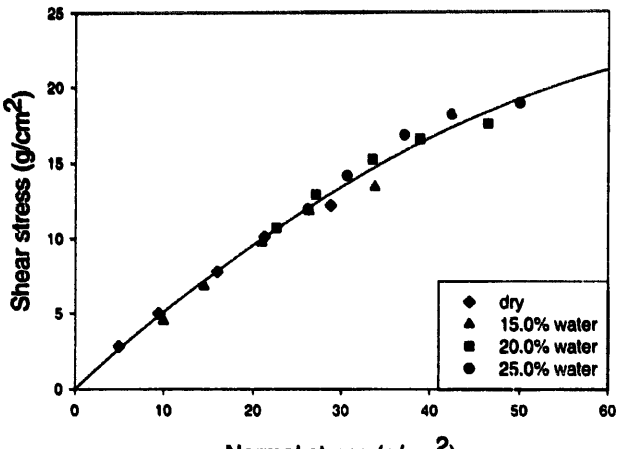

Fig. 15. Intrinsic yield locus of super D catalyst.

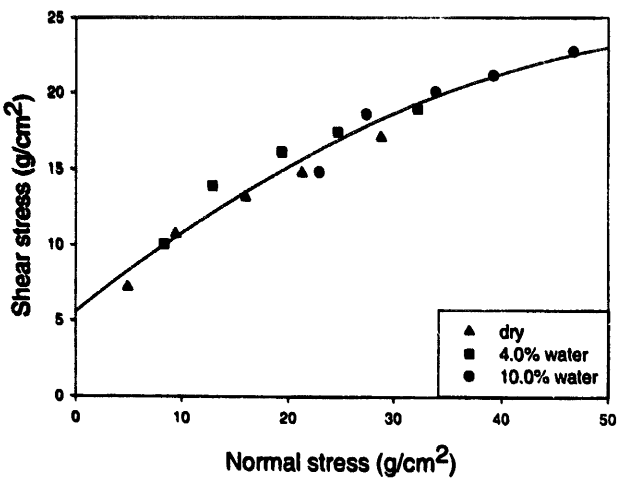

Fig. 16. Intrinsic yield locus of Leslie coal.

of the shift when an average value of ϕ is used in the computation. However, the same reasoning as for the straight yield locus applies. The Mohr circles shown in Fig. 1 represent the state of stress of the material at failure conditions from the point of view of an external observer. In the case of the dry powder, the state of stress is only due to the external consolidation applied on the specimen and the internal stresses are equal to the external stresses. The addition of moisture to the dry specimen creates an isotropic compressive stress in the powder. If the mechanical properties of the powder are not changed by the presence of moisture, the wet powder will yield under the same state of internal stress. This means that, from the point of view of an external observer, at failure conditions, the wet granular material is subjected to that state of stress minus the isotropic stress due to the presence of the pendular bridges. Or the principal stresses will be decreased by a constant amount equal to the isostatic stress, resulting in a horizontal shift of the original yield locus to the left. Again, the difficulty is that the estimation of the isostatic stress from the measured uniaxial stress is not as direct.

5. Conclusions

A simple theory was derived to reduce the number of tests to perform when investigating the effect of moisture on the flow properties of granular materials. It was assumed that the yield locus of a material at different moisture contents could be obtained with reasonable accuracy by horizontally shifting the original yield locus in the region of positive normal stresses by a value depending on the difference in tensile strengths.

The theory of shift enables one to determine the yield locus of a powder at any moisture content by shifting the original failure function in the region of positive normal stresses. For powders with straight yield loci, the shift is a function of the difference in tensile strengths of the material at the moisture content considered and the angle of internal friction. Shear tests on glass beads wetted with water and n-hexadecane showed that the curvature of the yield loci was small and, when Eq. (1) was used to calculate the shift, the results compared well with the measured horizontal separation distances. For crushed limestone, super D catalyst and Leslie coal wetted with water, the relation provided a good estimate of the shift. Eq. (1) requires the knowledge of the angle of internal friction ϕ and the tensile strength of the material. The angle of internal friction of the material is obtained from the yield locus. In the case of a slightly curved failure function, its value at a median normal stress can be calculated. Eq. (1) also implies that the angle of internal friction does not vary when the moisture content of the material is changed. This appears to be the case for the materials studied, glass beads, crushed limestone, super D catalyst and Leslie coal, powders that do not show tensile strength when dry but might be different for other materials. The effect of moisture on naturally cohesive materials needs to be studied.

6. Nomenclature

Roman notations

a Separation distance
d Particle diameter

h Vertical separation distance between the dry and the wet failure lines
n Shear index
T Tensile strength
ΔT Difference in tensile strength

Greek notations
δ Effective angle of internal friction
δ' Horizontal separation distance between the dry and the wet failure lines
Φ Angle of internal friction
$\Delta\sigma$ Isostatic yield
σ_1 Major principal stress
σ_2 Minor principal stress

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